

API Security with Tyk: Rate Limiting Algorithms

Tyk supports various types of algorithms. Discover how different algorithms help to smooth traffic spikes, enforce usage limits, and maintain API reliability under load.

Why Multiple Algorithms?

Not all API traffic are the same.

Examples:

- APIs get a **lot off traffic**
- APIs get **bursts**
- APIs need **fair sharing**
- APIs need to **stay simple and light**

Tyk offers various algorithms

- Distributed Rate Limited (DRL) - **Default**
- Redis Rate Limiter (RRL) - sliding window log
- Fixed Window Rate Limiter - basic & simple
- Redis Sentinel Rate Limiter - low latency + sliding log

Types of Rate Limiting

The Gateway implements the rate limiter, of which there are three types:

- Transactional
- Non-transactional
- Distributed

Default choice is Distributed, but this can be overridden through `tyk.conf`:

- Set `enable_redis_rolling_limiter` to true for transactional
- In addition, set `enable_non_transactional_rate_limiter` to true for non-transactional

Distributed Rate Limiter (DRL)

Think of this like having buckets on each gateway, and you divide the total allowed requests between them.

- Implemented using token bucket implementation that does not use redis
- Divides the configured rate limit between the number of gateway instances

Example:

- You set 100 requests per minute, and you have 2 gateways.
 - Each gateway gets 50 requests/minute.

If traffic is evenly distributed, this works great. But if one gateway gets more traffic than the other (due to sticky sessions or bad load balancing), it may rate limit early, even though the other gateway is idle.

Redis Rate Limiter (RRL)

This algorithm implements a sliding window log algorithm

- Spike arrest:
 - It won't allow traffic again until the rate falls below the limit for a full window
- Uses a lot of Redis CPU, since every request is logged.
- Keeps a log of every request in Redis, If too many requests come in over the set time, it blocks them

Example:

- You set 3000 requests/min.
 - If a client sends 4000 requests in one minute, they get blocked.
- Even if they slow down, the system waits until the whole minute passes with fewer than 3000 requests before allowing access again.

Fixed Window Rate Limiter

Fixed Window Rate Limiter will limit the number of requests in a particular window in time.

Once the defined rate limit has been reached, the requests will be blocked for the remainder of the configured window duration.

- Uses a simple counter per time window (e.g., per minute)
- Once the limit is hit, it blocks all further requests for that window

Example:

- Set 100 requests per minute
 - Client sends 100 quickly → they're blocked for the rest of the minute
 - New minute starts → they can send 100 again